M3 Topic 1 End of Topic Assessment · ANSWER KEY · QUICK (efficient method)

Algebra I · Module 3, Topic 1: Linear Equations and Inequalities · SE pp. 27–30 · 11 questions

Quick key

#	SE p.	Answer
1	27	**B. y = 20x + 50**
2	27	**J. No solution**
3	28	**D. s < −1**
4	28	**x = 1**
5	28	**Infinitely many solutions**
6	28	**y = 2.80x + 6.75**
7	29	**$y = -\frac{5}{3}x + 3$**
8	29	**$r = \frac{C}{2\pi}$**
9	29	**m < −4 (open circle at −4, shade left)**
10	30	**y = 2x + 11**
11	30	**n = 17**

Question 1

SE p. 27

Sofia: $50 from her aunt, saves $20 per week. Equation for amount y after x weeks?

B. y = 20x + 50

1. Rate (per week) = slope = 20; start amount = y-intercept = 50.
2. y = 20x + 50 → **B**.

Question 2

SE p. 27

What value of p makes 8 − 6(p + 4) = 12 − 2(3p + 10) true?

J. No solution

1. Distribute: 8 − 6p − 24 = 12 − 6p − 20.
2. Combine: −6p − 16 = −6p − 8.
3. Add 6p: −16 = −8 (false) → **J**.

Question 3

SE p. 28

Solution set for $4(2 - s) > -3s + 9$?

D. $s < -1$

1. $8 - 4s > -3s + 9$
2. Add $3s$: $8 - s > 9 \rightarrow -s > 1$.
3. Divide by -1, **flip**: $s < -1 \rightarrow$ **D**.

Question 4

SE p. 28

Solve $0.4(5x - 2) = -2.4x + 3.6$

$x = 1$

1. $2x - 0.8 = -2.4x + 3.6$
2. Add $2.4x$, add 0.8: $4.4x = 4.4$
3. $x = 1$

Question 5

SE p. 28

One solution, no solution, or infinite? $\frac{1}{3}(9x + 45) = 3(x + 3) + 6$

Infinitely many solutions

1. Left: $3x + 15$. Right: $3x + 9 + 6 = 3x + 15$.
2. Both sides identical ($0 = 0$) $\rightarrow$ **infinite**: every x works.

Question 6

SE p. 28

Chloe buys 3 eclairs at \$2.25 each and x doughnuts at \$2.80 each. Equation for total cost y?

$y = 2.80x + 6.75$

1. Eclairs: $3 \times 2.25 = 6.75$ (fixed).
2. Doughnuts: $2.80x$ (changes).
3. $y = 2.80x + 6.75$ (also accept $y = 6.75 + 2.8x$ or $y = 3(2.25) + 2.80x$).

Question 7

SE p. 29

Rewrite 5x + 3y = 9 in slope-intercept form.

$y = -\frac{5}{3}x + 3$

1. Subtract 5x: 3y = −5x + 9.
2. Divide by 3: $y = -\frac{5}{3}x + 3$.

Question 8

SE p. 29

C = 2πr. Solve for r.

$r = \frac{C}{2\pi}$

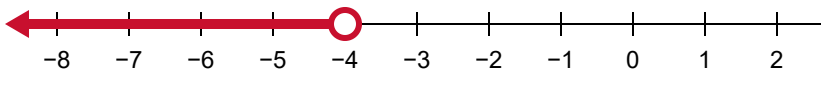
1. r is multiplied by 2π, so divide both sides by 2π.
2. $r = \frac{C}{2\pi}$

Question 9

SE p. 29

Solve 1.5m − 2 < −2(m + 8) and graph on the number line.

m < −4 (open circle at −4, shade left)

1. 1.5m − 2 < −2m − 16
2. Add 2m, add 2: 3.5m < −14
3. m < −4 (divided by positive → no flip)
4. 

Question 10

SE p. 30

Rewrite y − 5 = 2(x + 3) in slope-intercept form.

y = 2x + 11

1. y − 5 = 2x + 6
2. Add 5: y = 2x + 11

Question 11

SE p. 30

Triangle: height (2n − 4) in, base 8 in. Rectangle: length 6 in, width (3 + n) in. Areas equal. Find n.

n = 17

1. Triangle: $\frac{1}{2}(8)(2n - 4) = 8n - 16$. Rectangle: $6(3 + n) = 18 + 6n$.
2. $8n - 16 = 6n + 18 \rightarrow 2n = 34 \rightarrow n = 17$.
3. Check: both areas = 120 in².